

SHETH L.U.J & SIR M.V COLLEGE OF SCIENCE

SUBJECT : R-PROGRAMMING

PRACTICAL10:

AIM : Creating graphical reports using ,ggplot2 (R).

OUTPUT

The first screenshot shows the RStudio console with the following code and output:

```
R - R4.5.1 - ~/R
> library(ggplot2)
> library(dplyr)
> library(readr)
>
> # Load dataset
> data <- read_csv("health_lifestyle_dataset.csv")
Rows: 100000 Columns: 16
  Column specification
  _____
  Delimiter: ","
  chr (1): gender
  dbl (15): id, age, bmi, daily_steps, sleep_hours, water_intake_l, calories_consumed, smoker, alcohol...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
>
> # Peek at data
> head(data)
# A tibble: 6 x 16
   id age gender bmi daily_steps sleep_hours water_intake_l calories_consumed smoker alcohol
  <dbl> <dbl> <chr> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1     1  56 Male  20.5    4198         3.9         3.4        1602         0         0
2     2  69 Female 33.3   14359         9         4.7        2346         0         1
3     3  46 Male  31.6    1817         6.6         4.2        1643         0         1
4     4  32 Female 38.2   15772         3.6         2         2460         0         0
5     5  60 Female 33.6    6037         3.8         4.4        3756         0         1
6     6  25 Male  27.3   19495         5         4.4        1301         0         1

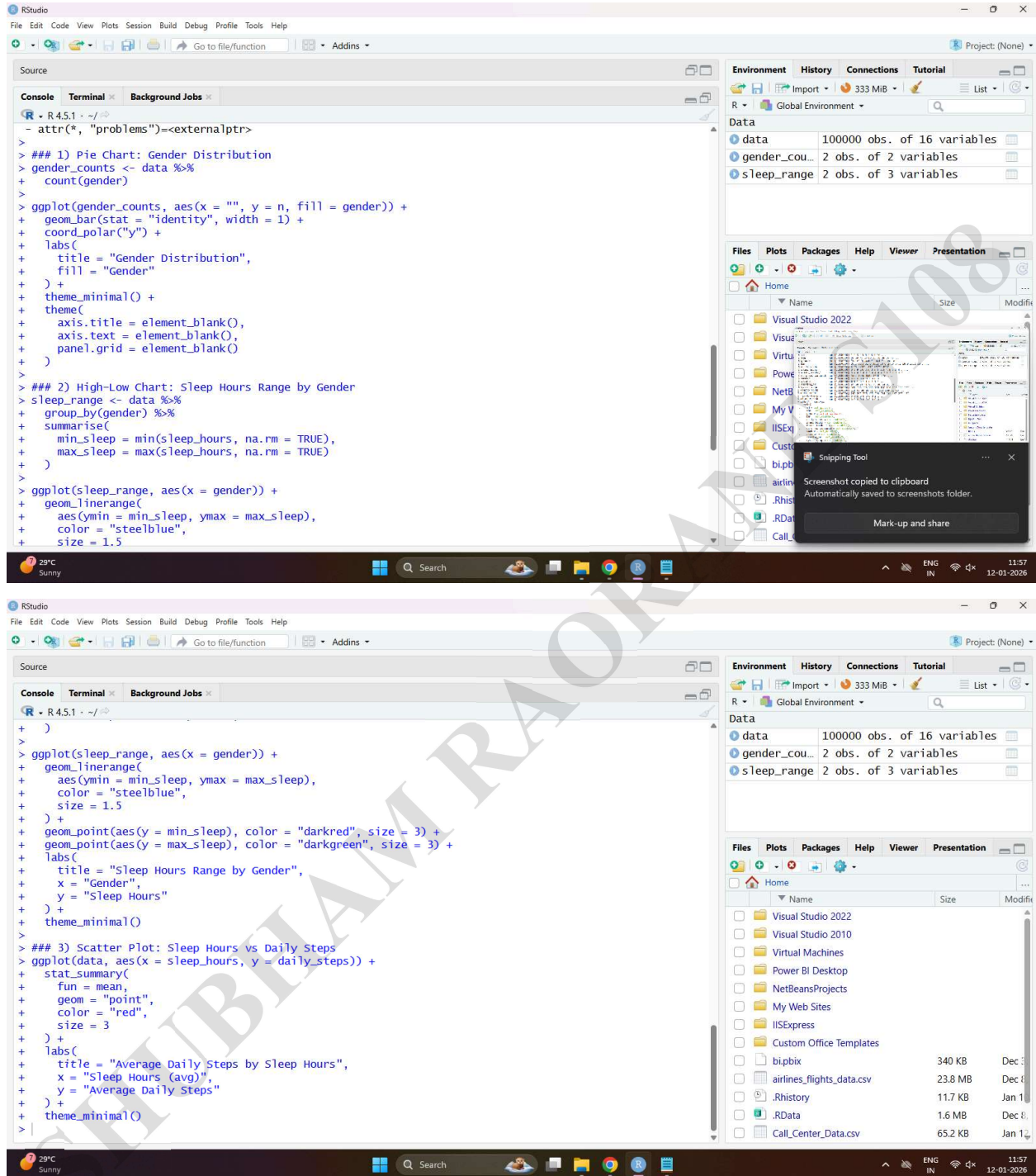
# i 6 more variables: resting_hr <dbl>, systolic_bp <dbl>, diastolic_bp <dbl>, cholesterol <dbl>,
#   family_history <dbl>, disease_risk <dbl>
> str(data)
'spc_tbl_ [100,000 x 16] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
 $ id      : num [1:100000] 1 2 3 4 5 6 7 8 9 10 ...
 $ age     : num [1:100000] 56 69 46 32 60 25 78 38 56 75 ...
 $ gender  : chr [1:100000] "Male" "Female" "Male" "Female" ...
 $ bmi     : num [1:100000] 20.5 33.3 31.6 38.2 33.6 27.3 37.1 18.9 18.2 23.5 ...
 $ daily_steps : num [1:100000] 4198 14359 1817 15772 6037 ...
 $ sleep_hours : num [1:100000] 3.9 9 6.6 3.6 3.8 5 9.5 4.8 5.1 4.5 ...
 $ water_intake_l : num [1:100000] 3.4 4.7 4.2 2 4 4.4 4.2 1.7 1.5 0.7 ...
 $ calories_consumed : num [1:100000] 1602 2346 1643 2460 3756 ...
 $ smoker   : num [1:100000] 0 0 0 0 0 1 0 0 0 ...
 $ alcohol  : num [1:100000] 0 1 1 0 1 1 0 1 1 ...
 $ resting_hr : num [1:100000] 97 68 90 71 98 73 90 64 91 54 ...
 $ systolic_bp : num [1:100000] 161 116 123 165 139 107 110 113 112 177 ...
 $ diastolic_bp : num [1:100000] 111 65 99 95 61 65 102 109 61 90 ...
 $ cholesterol : num [1:100000] 240 207 296 175 294 284 201 197 237 157 ...
 $ family_history : num [1:100000] 0 0 0 0 0 0 0 1 1 ...
 $ disease_risk : num [1:100000] 0 0 0 0 0 0 0 0 0 ...
 - attr(*, "spec")=
   .. cols(
   .. id = col_double(),
   .. age = col_double(),
   .. gender = col_character(),
   .. bmi = col_double(),
   .. daily_steps = col_double(),
   .. sleep_hours = col_double(),
   .. water_intake_l = col_double(),
   .. calories_consumed = col_double(),
   .. smoker = col_double(),
   .. alcohol = col_double(),
   .. resting_hr = col_double(),
   .. systolic_bp = col_double(),
   .. diastolic_bp = col_double(),
   .. cholesterol = col_double(),
   .. family_history = col_double(),
   .. disease_risk = col_double(),
   )
```

The second screenshot shows the RStudio environment pane with the following data:

Object	Size	Variables
data	100000 obs.	16 variables
gender_cou...	2 obs.	2 variables
sleep_range	2 obs.	3 variables

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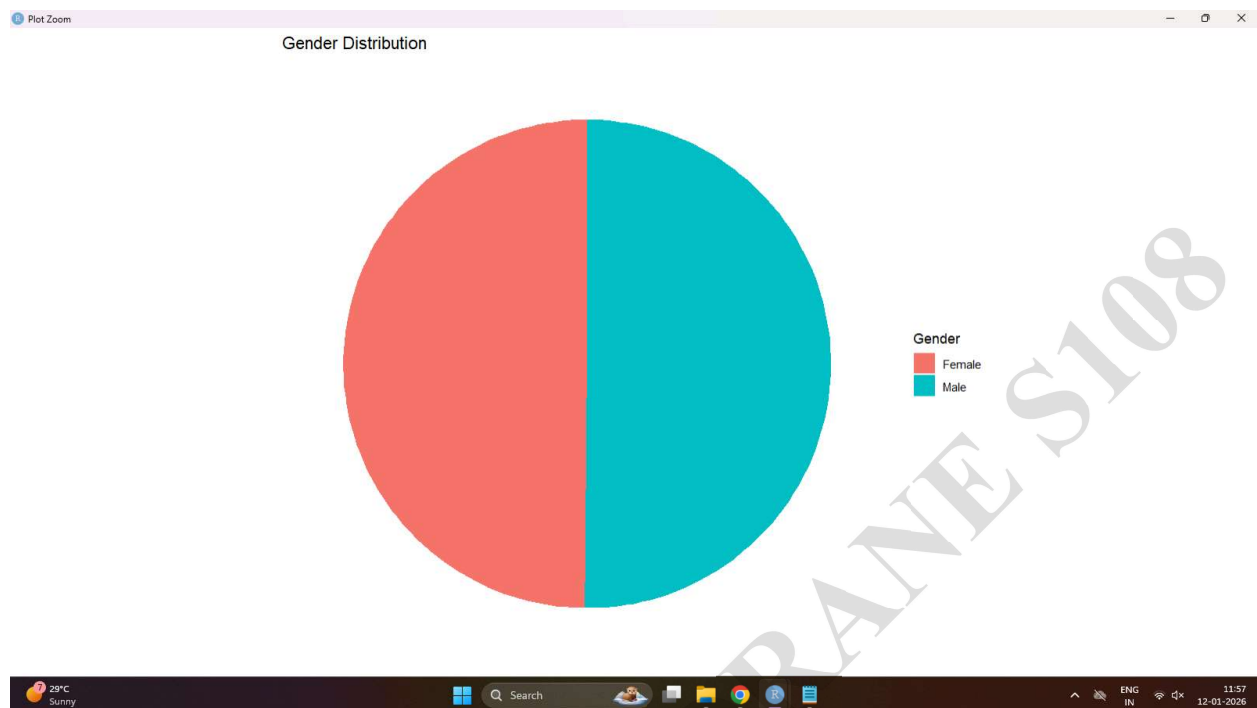
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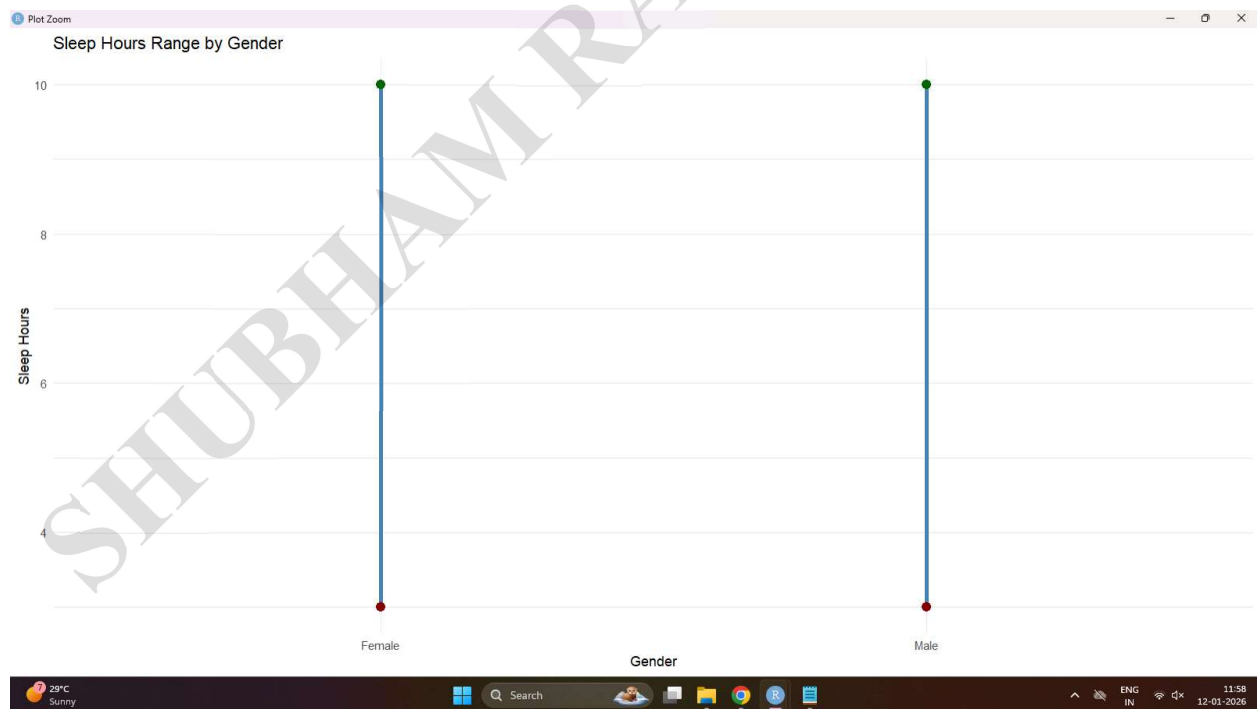
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PIE CHART: GENDER DISTRIBUTION



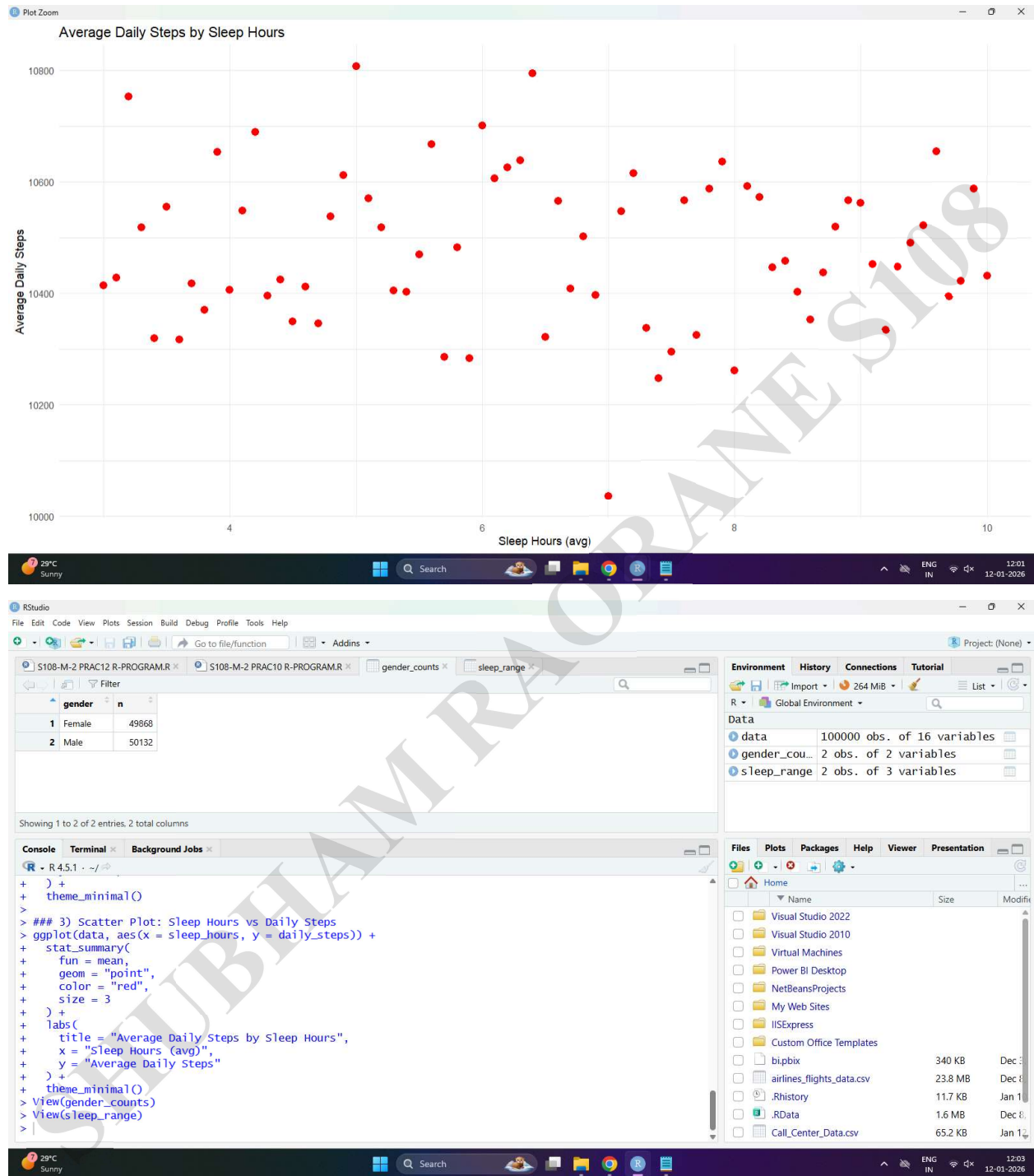
HIGH-LOW CHART: SLEEP HOURS RANGE BY GENDER



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SCATTER PLOT: AVERAGE DAILY STEPS BY SLEEP HOURS



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The screenshot shows the RStudio interface. The Environment pane on the right displays a data frame with 100,000 observations and 16 variables. The console shows the following R code:

```
R - R4.5.1 - ~/R
> ) +
+ theme_minimal()
>
> ### 3) Scatter Plot: Sleep Hours vs Daily Steps
> ggplot(data, aes(x = sleep_hours, y = daily_steps)) +
+ stat_summary(
+   fun = mean,
+   geom = "point",
+   color = "red",
+   size = 3
+ ) +
+ labs(
+   title = "Average Daily Steps by Sleep Hours",
+   x = "Sleep Hours (avg)",
+   y = "Average Daily Steps"
+ ) +
+ theme_minimal()
> View(gender_counts)
> View(sleep_range)
> |
```

PRACTICAL11:

AIM: Generating histograms and box plots using ggplot2 (R).

OUTPUT

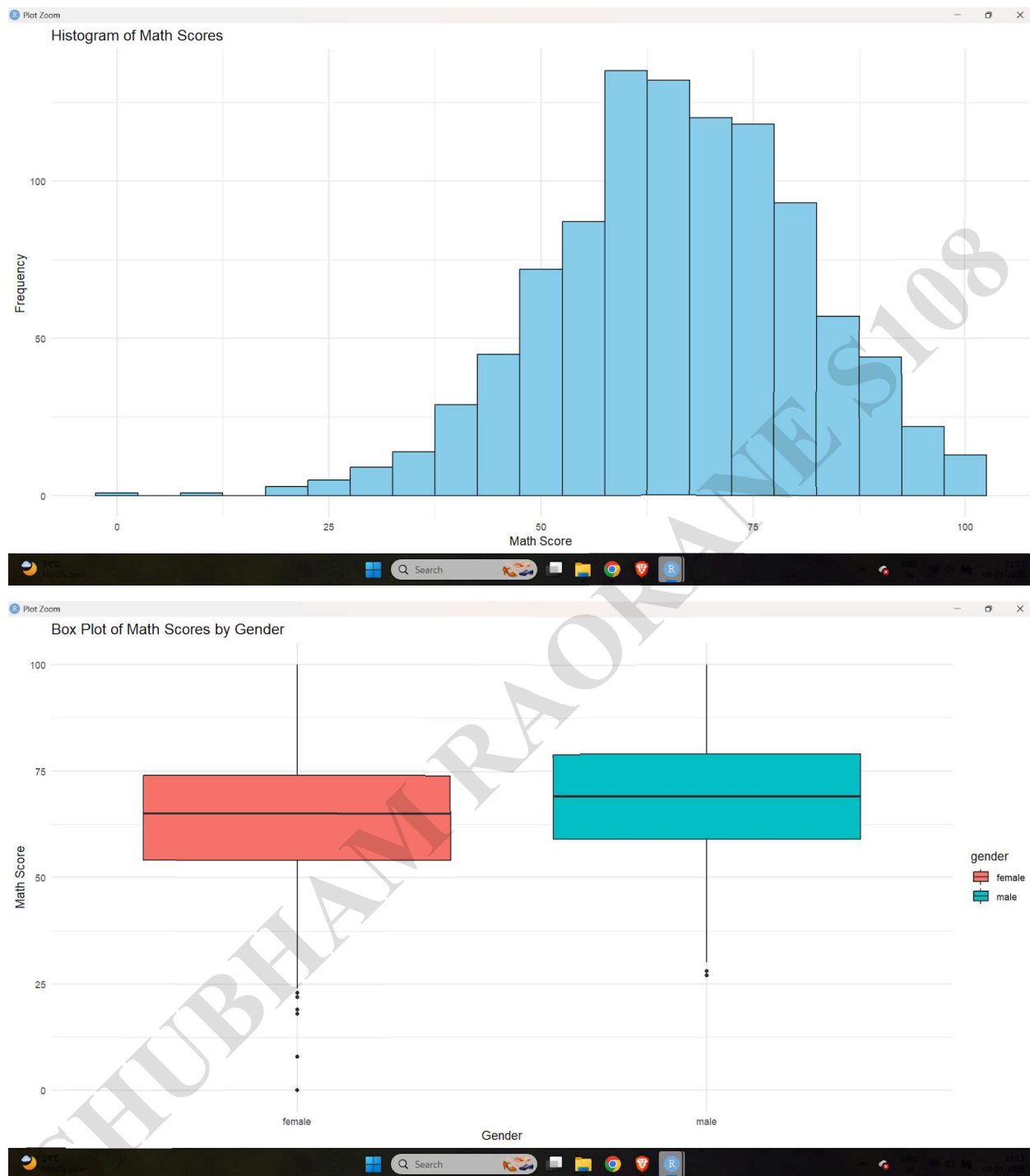
The screenshot shows the RStudio interface. The console shows the following R code:

```
R - R4.5.2 - ~/R
> library(ggplot2)
> library(readr)
>
> data <- read_csv("StudentsPerformance.csv")
Rows: 1000 Columns: 8
— Column specification
Delimiter: ","
chr (5): gender, race/ethnicity, parental level of education, lunch, test preparation course
dbl (3): math score, reading score, writing score

i use `spec()` to retrieve the full column specification for this data.
i specify the column types or set `show_col_types = FALSE` to quiet this message.
> ggplot(data, aes(x = math score)) +
+ geom_histogram(binwidth = 5, fill = "skyblue", color = "black") +
+ labs(
+   title = "Histogram of Math Scores",
+   x = "Math Score",
+   y = "Frequency"
+ ) +
+ theme_minimal()
> ggplot(data, aes(x = gender, y = `math score`, fill = gender)) +
+ geom_boxplot() +
+ labs(
+   title = "Box Plot of Math Scores by Gender",
+   x = "Gender",
+   y = "Math Score"
+ ) +
+ theme_minimal()
> |
```

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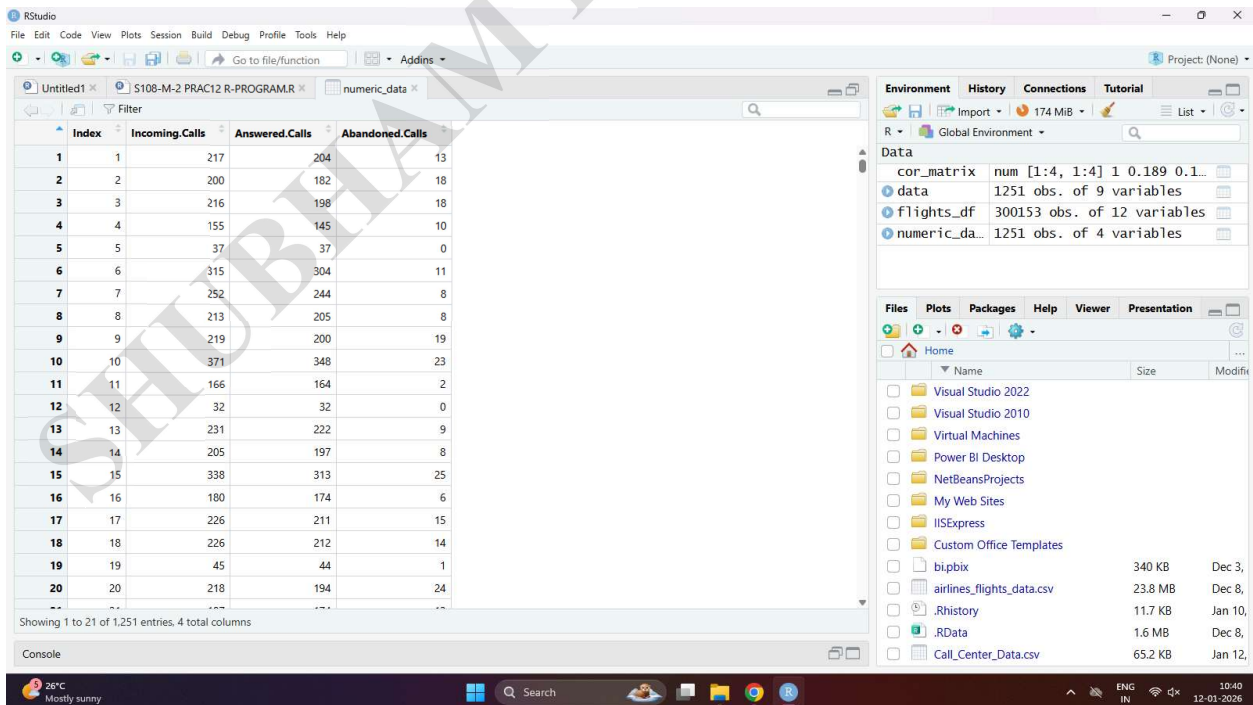
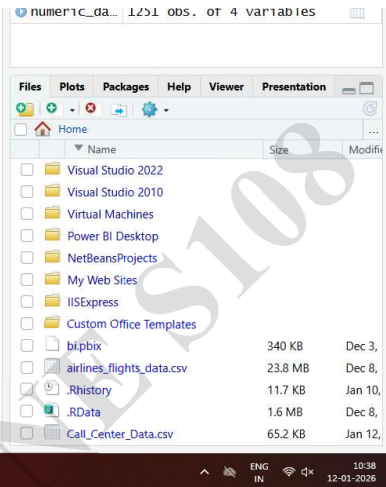
PRACTICAL12:

AIM: Generating correlation matrices using cor() (R).

OUTPUT

```
> getwd()
[1] "C:/Users/itlab/OneDrive/Documents"
> data <- read.csv("Call_Center_data.csv")
> numeric_data <- data[sapply(data, is.numeric)]
> cor_matrix <- cor(numeric_data)
> print(cor_matrix)
```

	Index	Incoming.Calls	Answered.Calls	Abandoned.Calls
Index	1.0000000	0.1888277	0.1900631	0.1271017
Incoming.Calls	0.1888277	1.0000000	0.9478892	0.7867473
Answered.Calls	0.1900631	0.9478892	1.0000000	0.5490854
Abandoned.Calls	0.1271017	0.7867473	0.5490854	1.0000000



	Index	Incoming.Calls	Answered.Calls	Abandoned.Calls
1	1	217	204	13
2	2	200	182	18
3	3	216	198	18
4	4	155	145	10
5	5	37	37	0
6	6	315	304	11
7	7	252	244	8
8	8	213	205	8
9	9	219	200	19
10	10	371	348	23
11	11	166	164	2
12	12	32	32	0
13	13	231	222	9
14	14	205	197	8
15	15	338	313	25
16	16	180	174	6
17	17	226	211	15
18	18	226	212	14
19	19	45	44	1
20	20	218	194	24

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The screenshot displays the RStudio environment. The main window shows a data frame with the following columns: Index, Incoming.Calls, Answered.Calls, Answer.Rate, Abandoned.Calls, Answer.Speed.AVG., Talk.Duration.AVG., and Waiting.Time.AVG. The data is presented in a table with 20 rows. The right-hand pane shows the Environment tab with a list of objects: cor_matrix, data, flights_df, and numeric_da. The Files pane on the right shows a directory structure with various folders and files, including Visual Studio 2022, Visual Studio 2010, Virtual Machines, Power BI Desktop, NetBeansProjects, My Web Sites, IISExpress, Custom Office Templates, bi.pbix, airlines_flights_data.csv, .Rhistory, .RData, and Call_Center_Data.csv. The bottom status bar shows the system clock as 10:40 on 12-01-2026.

Index	Incoming.Calls	Answered.Calls	Answer.Rate	Abandoned.Calls	Answer.Speed.AVG.	Talk.Duration.AVG.	Waiting.Time.AVG.
1	1	217	204	94.01%	13	0:00:17	0:02:14
2	2	200	182	91.00%	18	0:00:20	0:02:22
3	3	216	198	91.67%	18	0:00:18	0:02:38
4	4	155	145	93.55%	10	0:00:15	0:02:29
5	5	37	37	100.00%	0	0:00:03	0:02:06
6	6	315	304	96.51%	11	0:00:18	0:01:35
7	7	252	244	96.83%	8	0:00:13	0:01:50
8	8	213	205	96.24%	8	0:00:10	0:02:10
9	9	219	200	91.32%	19	0:00:15	0:02:18
10	10	371	348	93.80%	23	0:00:19	0:01:40
11	11	166	164	98.80%	2	0:00:12	0:02:10
12	12	32	32	100.00%	0	0:00:02	0:02:03
13	13	231	222	96.10%	9	0:00:08	0:01:49
14	14	205	197	96.10%	8	0:00:07	0:01:52
15	15	338	313	92.60%	25	0:00:13	0:01:51
16	16	180	174	96.67%	6	0:00:13	0:02:18
17	17	226	211	93.36%	15	0:00:20	0:02:04
18	18	226	212	93.81%	14	0:00:10	0:02:03
19	19	45	44	97.78%	1	0:00:02	0:02:04
20	20	218	194	88.99%	24	0:00:26	0:02:43